The Fundamental Theorem of Calculus

Section 1: FTC Part 1 (Differentiating Integrals)

1 Find $\frac{d}{dx}[\int_2^x \sqrt{t^3+1}\,dt]$.

a. $\sqrt{x^3+1}$

b. $\frac{3x^2}{2\sqrt{x^3+1}}$

c. $\sqrt{x^3+1}-3$

d. x^3+1

2 Find $\frac{d}{dx}[\int_0^{x^2} \cos(t)dt]$.

a. $2x\cos(x^2)$

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\sin(x^2)$

3 Find $\frac{d}{dx}[\int_{2x}^{3x} t^2 dt]$.

a. $19x^2$

b. $27x^2-8x^2$

c. $35x^2$

d. $5x^2$

Section 2: FTC Part 2 (Evaluation Theorem)

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4 Evaluate $\int_0^{\pi} \sin x dx$.

a. 2

b. 0

c. 1

d. -2

5 Evaluate $\int_1^4 \frac{1}{\sqrt{x}} dx$.

a. 2

b. 1

c. 4

d. $\frac{1}{2}$

6 Evaluate $\int_0^1 e^{2x} dx$.

a. $\frac{e^2-1}{2}$

b. e^2-1

c. $\frac{e^2}{2}$

d. $2(e^2-1)$

Section 3: Integration by u-Substitution

7 Evaluate $\int 2x(x^2+1)^4 dx$.

a. $\frac{(x^2+1)^5}{5}+C$

b. $\frac{(x^2+1)^5}{10}+C$

c. $(x^2+1)^5+C$

d. $\frac{2(x^2+1)^5}{5}+C$

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8 Evaluate the definite integral $\int_0^1 xe^{x^2}dx$.

a. $\frac{e-1}{2}$

b. $e-1$

c. $\frac{e}{2}$

d. $2(e-1)$

9 Evaluate $\int_0^{\pi/4} \tan x \sec^2 x dx$.

a. $\frac{1}{2}$

b. 1

c. $\frac{1}{4}$

d. $\frac{\sqrt{2}}{2}$

Section 4: Average Value of a Function

10 What is the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 3]$?

a. 6

b. 9

c. 18

d. 3

11 By the Mean Value Theorem for Integrals, what is guaranteed for a continuous function f on $[a, b]$?

a. There exists $c \in (a, b)$ such that $f(c) = f_{\text{avg}}$

b. $f'(c) = 0$

c. $f(a) = f(b)$

d. $f_{\text{avg}} = 0$

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12 What is the average value of $f(x) = \sin x$ on $[0, \pi]$?

a. $\frac{2}{\pi}$

b. $\frac{1}{\pi}$

c. 0

d. 1

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